

Alejandro S. Griffith — U.S. and Spain Citizen

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EDUCATION

Purdue University

West Lafayette, IN

Master of Science in Computer Science, concentration in Networking

August 2026 – May 2028

- Researching XR with a focus on networking, feedback, and low-latency improvements.

Bachelor of Science in Computer Science. GPA: 3.68

August 2022 – May 2026

- ACM Undergraduate Teaching Assistant Award, Dean's List (×8), Semester Honors (×7)

SKILLS AND TOOLS

Programming Languages: HTML, CSS, React, JavaScript, Java, C, C++, SQL, Python, Bash, and Assembly.

Tools: LaTeX, Terminal, GitHub, MySQL, SQLite, NodeJS, Neoj4, and MongoDB.

EXPERIENCE

Teaching Assistant

West Lafayette, IN

Purdue Computer Science

August 2024 - May 2026

- Courses: CS250 Computer Architecture; CS252 Systems Programming (×3)
- **Evaluate Student Work:** Evaluate student projects and lab assignments in C, C++, Shell Scripting, and Assembly.
- **Support Students:** Provide support by addressing student questions and problems during office hours or online forums.
- **Lead Lab Sessions:** Conduct lab sessions to help students with homework, exam preparation, and course-related questions.

Minecraft Developer

Jerez de la Frontera, Spain

Self-Employed

August 2016 - September 2020

- **Game Developer:** Designed and developed game add-ons in Java, and Shell Scripting (Bash), enhancing the gaming experience for hundreds of online players around the world. This role involved creating engaging content for Minecraft games, successfully generated revenue. Owned and managed servers using Shell scripting and Java that hosted over 10,000 registered users.

PROJECTS

Purdue Photography Club — Open-Source Photography Platform (github.com/PurduePhotographyClub),

Ongoing

- Built and maintain the open-source platform and internal tools supporting the Purdue Photography Club's website, Discord community, memberships, facilities, communications, and events.
- Developed across TypeScript, Astro, React, Cloudflare Workers, D1, R2, and Discord to support the club's digital community and operations.
- Technologies: TypeScript, Astro, React, Cloudflare Workers, D1, R2, Discord.

TCP Expo — Linux Kernel TCP Congestion Control (github.com/alesgsanudoo/TCP-EXPO), *March 2026*

- Designed and implemented a custom Linux kernel TCP congestion control algorithm that uses delay-based feedback to detect congestion before packet loss occurs.
- Developed adaptive congestion window control based on RTT inflation, including a lightweight negative-pulse reduction mechanism to reduce queue buildup while maintaining throughput and fairness.
- Evaluated TCP EXPO against CUBIC, Reno, and BBR using Flent experiments focused on throughput, latency, and fairness across single-flow and multi-flow network conditions.
- Technologies: C, Linux Kernel, Bash, Python, Flent.

EasyAccess — Desktop File Management (Team of 2) (github.com/alesgsanudoo/EasyAccess), *Aug 2025*

- Built a desktop application that provides a unified file management platform across three cloud storage services: Google Drive, OneDrive, and Dropbox—integrated with the local file system for seamless access.
- Developed the UI and provider integrations; implemented a Transfer Manager enabling chunked, concurrent uploads/downloads with automatic retries and cancel support.
- Technologies: TypeScript, Electron, Webpack, Google Drive API, Microsoft Graph (OneDrive), Dropbox API.

View more projects and repositories at www.alesgsanudoo.com/en/projects